

Professor: Mr. Daher

Cross-sectional data

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Tech Firm Leverage Analysis Report

Research Question:

Is there a relationship between a company's level of debt financing (Debt-to-Equity ratio) and its financial performance (stock return), its size (revenue), and its risk (volatility of returns)?

The dataset:

I extracted financial data from Yahoo Finance for 400 publicly listed companies from the S&P 500, using the closing date of December 31, 2024. For the rate of return (average daily return annualized) and risk (standard deviation of daily returns annualized), I used daily stock prices over the entire year 2024. The dataset was downloaded automatically using a Python script with the yfinance library (see “How I create raw CSV file.ipynb”).

I restricted the sample to companies operating in the same industry (technology). This ensures better comparability since the Debt-to-Equity ratio varies significantly across industries. Keeping companies from a single sector allows me to interpret the relationship between leverage (debt financing), performance, and risk more consistently.

I selected five variables for our dataset:

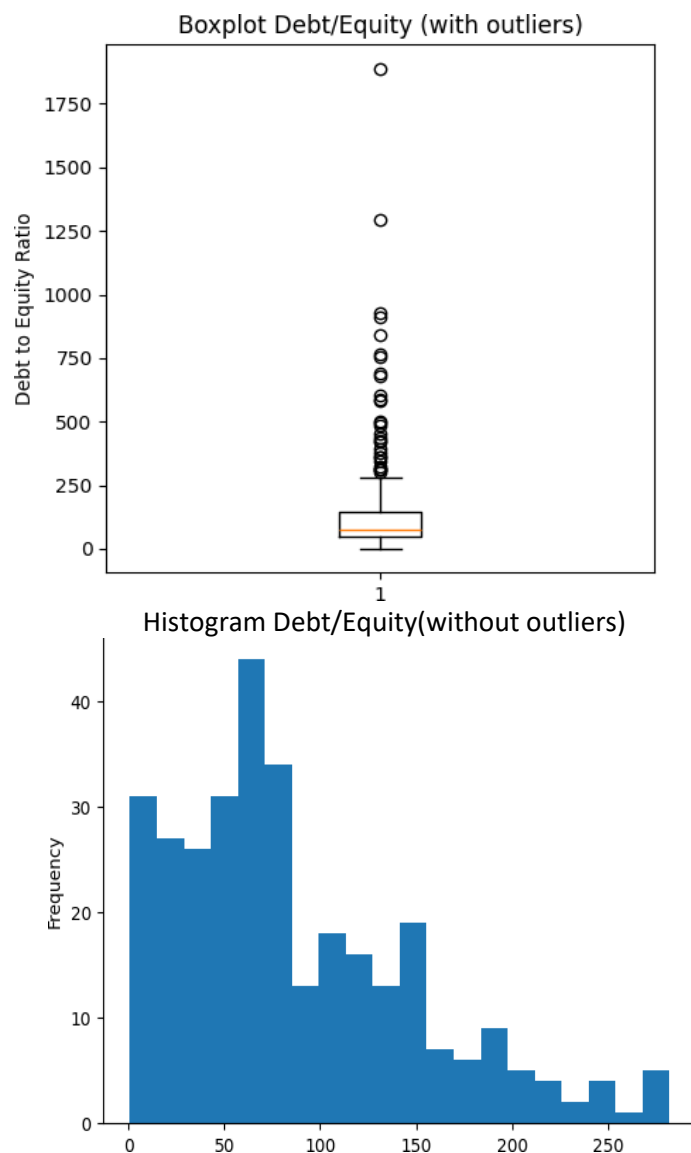
- Stock Ticker
→Identifies each publicly listed company.
- Long-term Debt-to-Equity Ratio (%)
→Measures the extent to which a company finances its activities through debt relative to equity. A value above 100% indicates more debt than equity, while a value below 100% indicates more equity than debt.
- Risk (Standard Deviation of Returns)
→Represents the volatility of the stock's daily returns over 2024. Higher values indicate a riskier stock.
- Rate of Return (Annualized)
→Measures the profitability of the stock over the year 2024.
- Revenue
→Used as an indicator of the company's size (larger firms typically report higher revenue).

Data cleaning:

Because the dataset contains many observations, I first removed incomplete entries using `dropna()`. This ensures consistency in the correlation analysis, as missing values could distort the results. Next, I assessed the presence of outliers using a boxplot, and applied the Interquartile Range (IQR) method:

Outliers = values outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$

Removing outliers prevents extreme values from disproportionately influencing the visualizations and correlations. This allowed me to produce clearer scatter plots and more reliable insights.



Data Manipulation and Transformation:

The dataset was extracted from Yahoo Finance specifically for the purpose of answering our research question. Because the dataset was built with only the required variables from the start, no variable needed to be filtered out. Each variable directly contributes to the analysis and helps measure the relationship between leverage (Debt-to-Equity), performance (Return), firm size (Revenue), and risk (Volatility).

Core statistics:

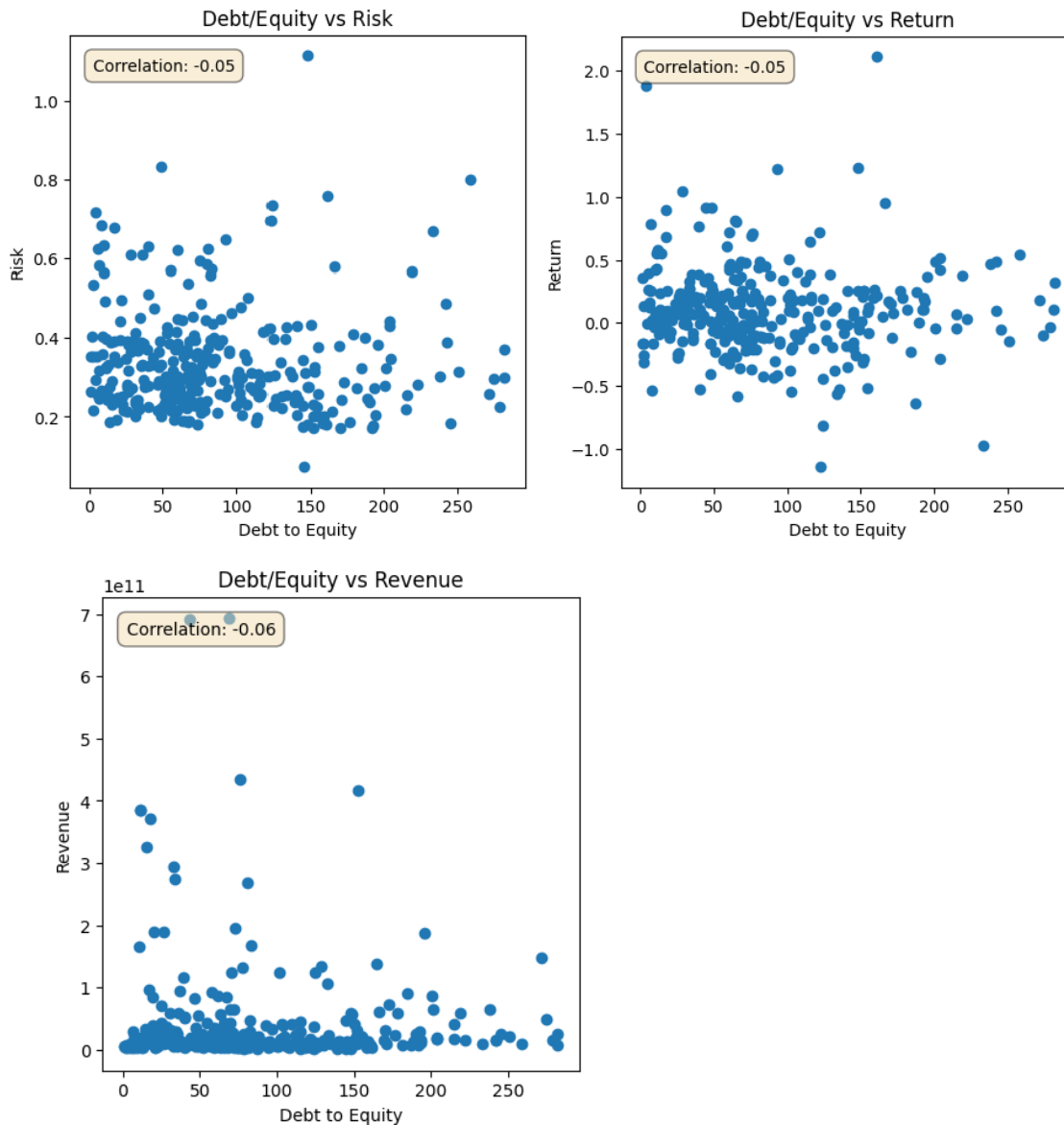
	Debt_to_Equity	Risk	Return	Revenue
count	344	344	344	3.440000e+02
mean	129.164189	0.336358	0.095321	3.928071e+10
std	181.623974	0.134917	0.359176	8.062123e+10
min	0.887000	0.072489	-1.143474	9.405610e+08
25%	45.656000	0.244614	-0.108462	7.490325e+09
50%	75.621000	0.302432	0.063679	1.436450e+10
75%	146.734500	0.388337	0.239145	3.237450e+10
max	1888.690000	1.114902	2.111721	6.931530e+11

Data Visualization:

I produced several visualizations to better understand the dataset and support our analysis:

- A box plot of the Debt-to-Equity ratio to identify and justify the removal of outliers.
- A histogram (distribution plot) of the Debt-to-Equity ratio without outliers to show a clearer representation of how firms are distributed in terms of leverage.
- Three scatter plots showing the relationship between the Debt-to-Equity ratio and each of our other variables:
 - Revenue (firm size),
 - Risk (volatility of returns),
 - Rate of Return (profitability).

Each scatter plot includes the correlation value on the top left corner of the graph.



Key Results:

Our analysis shows no significant correlation between the Debt-to-Equity ratio and any of the three variables:

Revenue (company size), Risk (volatility), or Rate of Return (profitability).

These findings suggest the following interpretations: A company does not systematically use more debt as it grows: firm size is not related to leverage decisions.

A company's stock performance (rate of return) is not directly influenced by whether the firm relies more on debt or equity financing.

A company's stock risk (volatility of returns) is not determined by its level of debt financing.

Our Recommendations:

When selecting a company to invest in, do not base your decision on the assumption that its Debt-to-Equity ratio makes the stock safer or more profitable. Our analysis shows that leverage (using more debt versus equity) is not linked to stock performance, stock risk, or company size. Similarly, do not assume that large companies systematically use more debt or that smaller ones rely more on equity; our results show no consistent pattern.

Challenges:

At first, I believed that finding no correlation between our variables meant that our analysis was not meaningful. I even considered changing variables to obtain a “more interesting” result. However, after further reflection, I realized that our research question was legitimate, and that the absence of correlation is an insight in itself. Demonstrating that leverage does not explain stock performance, risk, or size is just as valuable as finding a strong correlation.

A second challenge was technical. In our first code iteration, I extracted the data directly from Yahoo Finance in the same notebook used for data cleaning and visualization. Because this required downloading data for 400 companies and 4 variables each time the code ran, the execution was very long and heavy. To solve this, I split the work into two separate codes:

“How I create raw CSV file.ipynb” → Extracts stock data from Yahoo Finance and saves it as a CSV file.

“DHS4_Script_Bender.ipynb” → Loads the CSV, cleans the data, performs transformations, and generates visualizations.

This approach significantly reduced computation time and made the workflow much more efficient. Finally, I discovered that the dataset contained many extreme outliers, especially in the Debt-to-Equity ratio. These values compressed all the other points in the scatter plots, making the visualizations unclear. To address this, I removed the outliers using the $1.5 \times \text{IQR}$ method, which allowed me to generate cleaner and more interpretable plots.

AI Use Disclosure Statement

Tools used(name&version): Gemini, google colab

Where used (cells/sections): lines 44 to 53

(AI-tagged as # [AI])

Purposes (brief): used to add the correlation value in text on the scatter plots

I confirm that all outputs were reviewed, edited, and verified; all results are reproducible from the submitted notebook and CSV; and each presenter can orally defend the submitted work. I accept responsibility for any errors.

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